

BERT Sentiment Analysis on IMDb Movie Reviews

Final Report - DATASCI 224 AI Spring 2026

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1. Problem and Dataset

This project focuses on binary sentiment classification using movie reviews from the IMDb dataset. Each review is labeled as either positive or negative, and the goal is to train a model that can accurately predict the sentiment of reviews it has not seen before. Sentiment analysis is a common task in natural language processing because it can be used to understand opinions, customer feedback, and user-generated content at scale.

For this project, a subset of the IMDb dataset was used to keep training and evaluation manageable. The training set contains 4,000 reviews, evenly split between positive and negative sentiment, while the test set contains 2,000 reviews with the same balanced class distribution. The original IMDb dataset is organized into separate folders for positive and negative reviews in both the training and testing sets, making it straightforward to load the text and assign labels.

Working with a balanced dataset helps reduce the risk of the model favoring one class over the other during training. Since the dataset contains equal numbers of positive and negative reviews, evaluation metrics such as accuracy, precision, recall, and F1-score provide a more reliable picture of how well the model performs. A summary of the dataset used in this project is shown in Table 1, and the class distribution is illustrated in Figure 1.

Table 1

IMDb Movie Review Dataset Summary

Dataset item	Value
Task	Binary sentiment classification
Training set used in notebook	4,000 reviews total
Test set used in notebook	2,000 reviews total
Class balance (train)	2,000 positive / 2,000 negative
Class balance (test)	1,000 positive / 1,000 negative
Input format	Raw review text files
Sequence length used	256 tokens (truncated / padded)

Table 1. Summary of the IMDb movie review dataset used for sentiment classification, including dataset size, class distribution, input format, and model sequence length. The dataset contains balanced positive and negative review classes for both training and testing splits

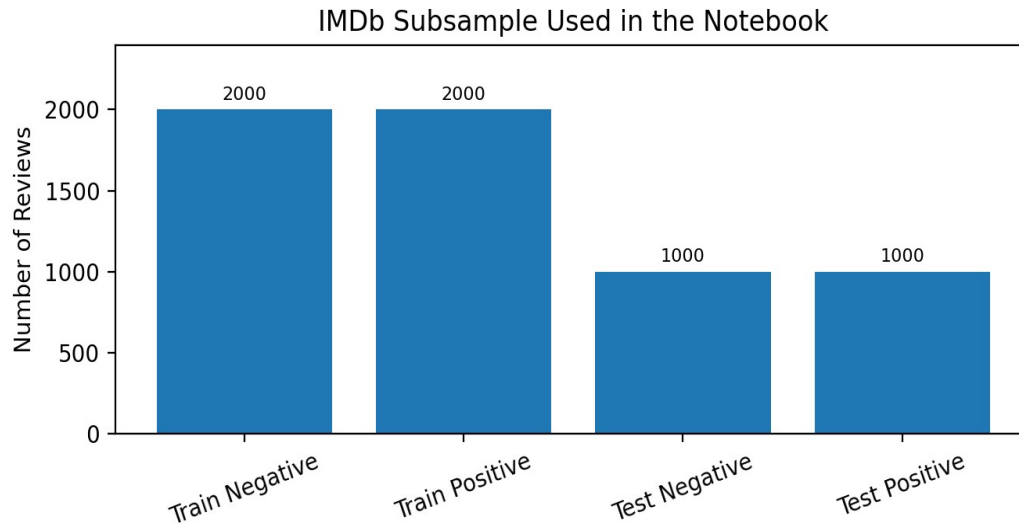


Figure 1. IMDb Subsample Used in the Notebook. *The subsample is perfectly balanced across both classes in both splits, which reduces label-skew issues during fine-tuning.*

2. Machine Learning Algorithm: BERT Fine-Tuning

The model used in this project is BERT (Bidirectional Encoder Representations from Transformers). The notebook uses the pretrained bert-base-uncased model and fine-tunes it on IMDb movie reviews to classify sentiment as positive or negative.

BERT works best when meaning depends on context rather than individual words. Movie reviews are a good example because sentiment is often expressed through phrases, comparisons, and negation. A phrase such as "not bad" would be difficult to interpret using simple word counts, but BERT can use the surrounding context to determine the intended meaning. The model also assumes that the language patterns it learned during pretraining remain useful for the task being fine-tuned. Because BERT was trained on a large amount of text before being applied to this project, it already has a general understanding of language structure and word relationships. Fine-tuning adjusts that knowledge for sentiment classification instead of learning everything from scratch. BERT tends to perform well when there is enough labeled data for fine-tuning and when understanding context is important for making predictions. It is especially effective for text classification tasks such as sentiment analysis, where the meaning of a review often depends on how words are used together rather than on the presence of specific keywords alone.

At the core of BERT is the self-attention mechanism:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{sqrt(d_k)}\right)V$$

Self-attention allows the model to weigh the importance of different words when processing a sentence. Instead of reading text one word at a time, BERT can consider relationships between words across the entire review.

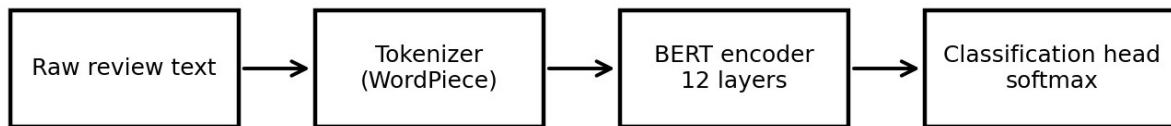
Each review is first broken into WordPiece tokens and converted into numerical embeddings. These embeddings pass through multiple Transformer layers that apply self-attention and feed-forward operations. After the final layer, the representation associated with the special [CLS] token is used as a summary of the review. A classification layer then predicts the probability that the review is positive or negative:

$$p(y|x) = \text{softmax}(Wh_{CLS} + b)$$

The model is trained by minimizing cross-entropy loss:

$$L = -\sum y_i \log(p_i)$$

In sum, the review is first split into WordPiece tokens and converted into embeddings that BERT can process. Those embeddings pass through a series of Transformer layers, where the model learns which words matter most and how they relate to one another in the review. After the final layer, BERT produces a representation for the special [CLS] token, which is used as a summary of the entire review. That representation is passed to a classification layer that predicts whether the sentiment is positive or negative. During training, the model updates its weights using cross-entropy loss so that its predictions become more accurate over time.



The [CLS] representation is passed to a linear layer for sentiment prediction.

Figure 2: BERT Sentiment Analysis Pipeline.

The notebook uses AdamW with a learning rate of 2e-5, batch size 16, 2 training epochs, and a maximum sequence length of 256 tokens. These settings are consistent with common BERT fine-tuning practice: a small learning rate preserves useful pretrained weights, while short fine-tuning reduces overfitting.

3. Results and Discussion

Table 2
Test Set Performance

Metric	Value
Accuracy	90.05%
Precision	87.75%
Recall	93.10%
F1-Score	90.34%

Table 2. Performance metrics for the fine-tuned BERT model on the IMDb test dataset.

Table 3
Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.93	0.87	0.90	1000
Positive	0.88	0.93	0.90	1000

Table 3. Classification results for the positive and negative sentiment classes on the IMDb test dataset.

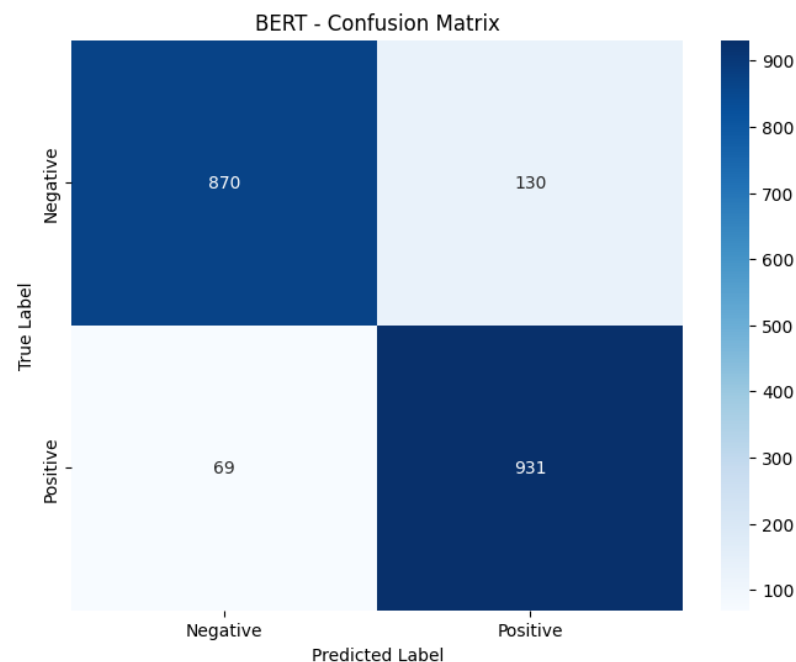


Figure 3. Confusion Matrix for the Fine-Tuned BERT Model. The confusion matrix summarizes classification outcomes on the IMDb test dataset. Most predictions fall along the diagonal, indicating correct classifications, while relatively few reviews are misclassified.

Discussion: The model reached 90.05% accuracy on the test set, with an F1-score of 90.34% (Table 2). Overall, the results were consistent across the evaluation metrics, suggesting that the model handled unseen reviews reasonably well. The gap between precision (87.75%) and recall (93.10%) shows that the model was slightly better at finding positive reviews than avoiding false positives, but the difference was not large.

The class-level results in Table 3 tell a similar story. Positive reviews were identified somewhat more reliably than negative reviews, with recall scores of 93% and 87%, respectively. Even so, both classes ended up with nearly identical F1-scores of about 0.90. The balanced class distribution shown in Figure 1 likely helped prevent the model from leaning too heavily toward one sentiment class during training.

Figure 3 shows the confusion matrix for the final model. Most predictions fall along the diagonal, which means the reviews were classified correctly. Out of 1,000 negative reviews, 870 were classified correctly and 130 were predicted as positive. For positive reviews, 931 out of 1,000 were classified correctly. These results line up with the higher recall observed for the positive class.

One reason the model performs well is that sentiment often depends on context. As shown in Figure 2, reviews pass through several Transformer layers before classification. This allows BERT to consider how words relate to one another instead of treating them as isolated terms. That matters in movie reviews, where sentiment is often expressed through phrases, comparisons, or negation rather than single keywords.

References:

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